

NIKOS CHRYSOGELOS

Electrical and Computer Engineer

📅 10 March 1990 @ nkchrys@gmail.com ☎ +41 76 687 76 94
in Nikos 📍 Zürich, Switzerland 🔗 nkchrys 🌐 Nikos' web



WORK EXPERIENCE

Anybotics Electrical Engineer

📅 Jun 2024 – Ongoing 📍 Zürich, CH

- **Senior engineer** in the new custom battery design of ANYmal X
- **Senior engineer** in the latest product iteration: custom GPU motherboard design (PCIe, USB 3, GMSL, MIPI)
- **Expert engineer** in EMC testing
- **Lead engineer** in wireless topics (Wi-Fi, Cellular, RF antenna design)

Fotokite Electrical Engineer

📅 Jan 2018 – April 2024 📍 Zürich, CH

- **Lead engineer** in manufacturing and scaling production from 10 to 300 in USA and Switzerland.
- From prototype to **CE and FCC certifications** by developing an in-house EMC lab to iterate faster and improving the electrical design accordingly
- **Expert engineer** in power electronics design and Power Line Communication
- **Lead engineer** in camera, WiFi and GPU modules integrated designs
- Hands-on hardware testing, debugging, soldering, fast prototyping, and firmware development

ABB, CRC Power Electronics Engineer

📅 Sep 2016 – Feb 2017 📍 Baden, CH

Analysis and comparison of **control algorithms** for **grid-connected power converters**.

ETH Zürich Research Assistant

📅 May 2016 – Sep 2016 📍 Zürich, CH

Development of an efficient **DC micro-grid** for a lighting installation.

CERN Electronics Engineer

📅 Feb 2013 – Apr 2014 📍 Geneva, CH

Designing and developing **control systems** and **wireless sensor network** for the new cooling plant of the CMS experiment using PLC and AVR.

JOB PHILOSOPHY

Passionate about **revolutionary** hardware and **human-centered** products. Excited to be working with inspired and motivated colleagues who are ready to create **meaningful** and **sustainable** innovation.

FAVOURITE PROJECTS

 **Unleash Rwanda**
Development of off-grid refrigeration systems for remote areas in Uganda.

 **Orama**
Most innovative idea in **PowerUp!2019** organized by EIT InnoEnergy, Greece. Data analytics for load forecasting in neighborhood-level electricity grid.

 **SolarBites**
Electric cooking for refugee camps using solar panels and flexible PCBAs, **Hackathon 2019, Cern**.

STRENGTHS

Altium Designer

Manufacturing

EMC design

Power systems

Digital electronics

High speed design

Power electronics

Resonant converters

Battery management

Ethernet

WiFi

Power Line Communication

MIPI

CAN

PCIe

USB

Flex PCBA

Linux

Matlab

Simulations

Python

Git

PLC-Siemens

Drones

ARM Cortex-M

Firmware

C

C++

Energy data analytics

Smart meters

Jira

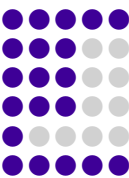
Notion

Confluence

Wrike

LANGUAGES

English
French
German
Spanish
Portuguese
Greek



EDUCATION AND PUBLICATIONS

ETH Zürich MSc - Power electronics and energy systems

 Sep 2015 – Sep 2017  Zurich, CH 5.6/6

- "An 380V / 48V, 1.5 kW Partial-Power Precontrolled DC Transformer", D. Neumayr, M. Voehringer, N. Chrysogelos, G. Deboy and J.W. Kolar, IEEE Power Electronics Transactions 2018, pp. 1-13.
- "Identification of Consumer Behavior via Smart Meters: Novel Artificial Intelligence and Machine Learning Disaggregation Algorithms ", Nikolaos Chrysogelos, Chin Jun Xing , Prof. Dr. Gabriella Hug.
- "Energy Storage for Mechatronic Energy Harvesting System", Nikolaos Chrysogelos, Michael Flankl, Prof. Dr. Johann Walter Kolar.



NTUA Diploma - Computer engineering

 Sep 2007 – May 2013  Athens, GR 8.1/10 (20%)

- "Assessment of the reverse operational characteristics of SiC JFETs in a diode-less inverter", G. Kampitsis, P. Stefas, N. Chrysogelos, S. Papathanasiou and S. Manias, Industrial Electronics Society, IECON 2013 - 39th Annual Conference of the IEEE, Vienna, 2013, pp. 477-482.
- "Performance consideration of an AC coupled gate drive circuit with forward bias for normally-on SiC JFETs", G. Kampitsis, S. Papathanasiou and S. Manias, Energy Conversion Congress and Exposition (ECCE), 2013 IEEE, Denver, CO, 2013, pp. 3224-3229.

INTERESTS AND ACTIVITIES

- Participation in movements and NGOs for successful integration of refugees and asylum seekers coming from war zones into Europe (ETH Discovery Semester, Hackathon - Cern, Programm des Kantons Zürich).
- Traveling as a volunteer in NGOs in Nepal and Kenya (Wheeling 2 help, Conscious impact Nepal, Sadhana forest Kenya).
- Volunteer in the first competition for people with disabilities using advanced robotic technologies, Cybathlon 2016 in Switzerland.
- Official underground and exhibition guide at CMS-ATLAS and Computer Centre at Cern.
- Humanitarian Hackathon participant 2018-2020, co-organized by Cern
- Football, basketball, snowboarding, half-marathon runner, theatre.

REFEREES

Dr. Sergei Lupashin

@ contact details available upon request

✉ @Founder at Fotokite

Dr. Mike Hamer

@ contact details available upon request

✉ @ex CTO at Fotokite

Karl Schwabe

@ contact details available upon request

✉ Systems Engineer Lead at Fotokite

AWARDS

- Bodossaki Foundation, MSc studies.
- Kiprianidis Foundation, Diploma studies.
- High-school performance awards by mayor of Athens.
- Gold medal in school basketball competition.
- Award for the project "Cultural and historical characteristics of our hometown (Kolonos , Athens)".

WORKSHOPS

- "Conflict management for peaceful dialogue" at Le Fai, Erasmus+, France 2017.
- "The Control Systems of the LHC", workshop at CERN, Switzerland 2014.
- "From Algorithms to implementation in FPGA and Complex System on Chip design" at University of Las Palmas, EESTEC student organization, Spain 2013.
- "Smart Energy for a better world", EESTEC student organization, ABB CRC Krakow Poland 2012.
- "Empowering youth possibilities in Europe", Erasmus+, Turkey 2012.